COMPUTER SYSTEMS FUNDAMENTALS
(4COSC004W)

Week 1. Part 1 of 2

THE NATURE OF NUMBERS

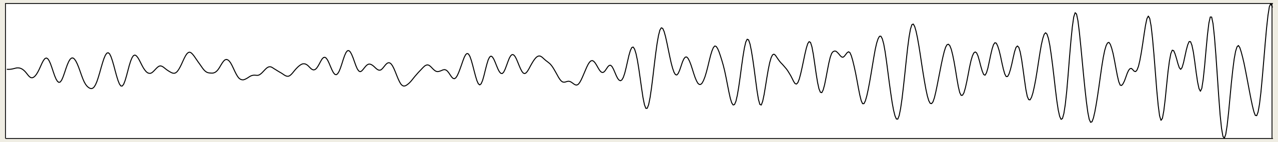
Positional number systems

By the end of this part, you will:

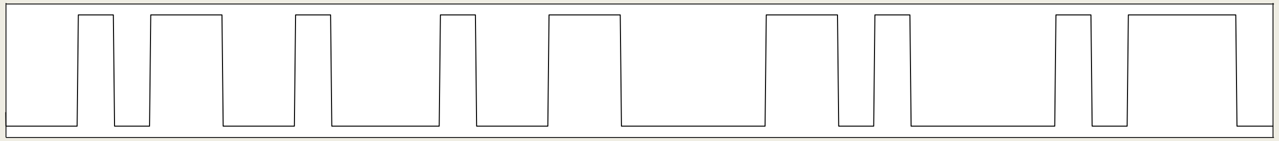
- Understand the concept of Positional Number Systems
- Be able to count and interpret natural numbers
  - *Positive (unsigned) Integers*
- Understand the following number systems:
  - *Decimal / Denary – Base 10*
  - *Binary - Base 2*
- Be able to count in Binary

Binary : Why use Binary?

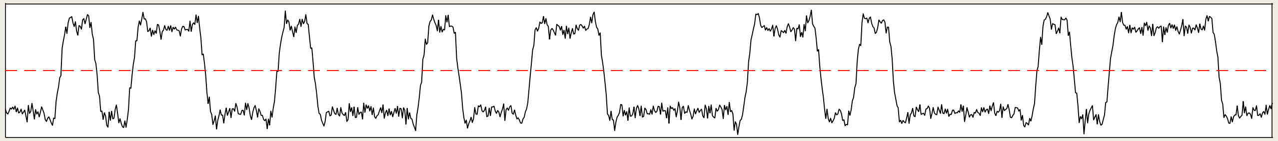
- Computer Systems use Binary to represent data (logic-0 and logic-1)
- Binary coding to code real-life signals



An example to an Analog Waveform



Binary coded Waveform to represent digital information.



Binary coded Waveform with Noise. The code can still be acquired despite the distortion added.

- Binary is used for Digital systems
  - *On/Off*

Decimal / Denary – Base 10

1	0

Decimal / Denary – Base 10

1 0

10 1

Decimal / Denary – Base 10

1	0
10^1	10^0
10	1

Decimal / Denary – Base 10

2	1	3
10^2	10^1	10^0
100	10	1

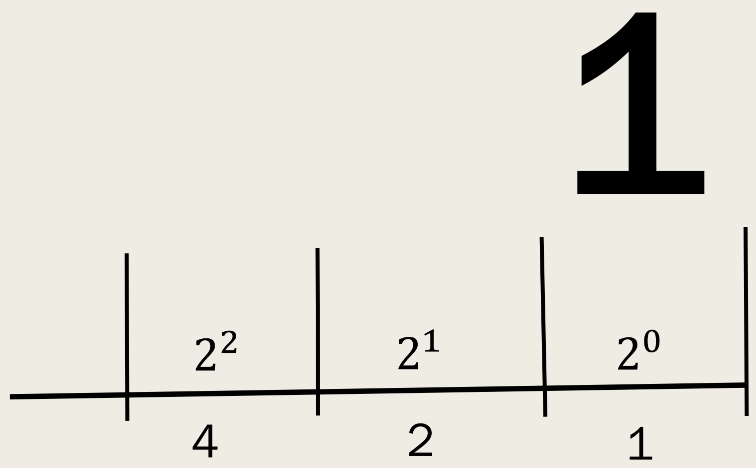
Base 2
Binary

		0
2^2	2^1	2^0
4	2	1

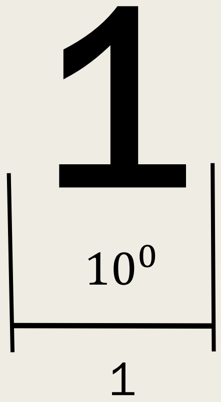
Base 10
Denary

0
10^0
1

Base 2
Binary



Base 10
Denary



Base 2
Binary

1 0

2^2	2^1	2^0
4	2	1

Base 10
Denary

2

10^0
1

Base 2
Binary

1 1

	2^2	2^1	2^0
	4	2	1

Base 10
Denary

3

10^0
1

Base 2
Binary

1 0 0

2^2	2^1	2^0
4	2	1

Base 10
Denary

4

10^0
1

Base 2
Binary

1	0	1
2^2	2^1	2^0
4	2	1

Base 10
Denary

5
10^0
1

Base 2
Binary

1	1	0
2^2	2^1	2^0
4	2	1

Base 10
Denary

6
10^0
1

Base 2
Binary

1	1	1
2^2	2^1	2^0
4	2	1

Base 10
Denary

7
10^0
1

Base 2
Binary

1	0	0	0
2^3	2^2	2^1	2^0
8	4	2	1

Base 10
Denary

8
10^0
1

An exercise for you:

Convert the 4-bit **Binary** value **1100** into **Denary:**

Base 2
Binary

1 1 0 0

2^3	2^2	2^1	2^0
8	4	2	1

Base 10
Denary

8 + 4 = 12

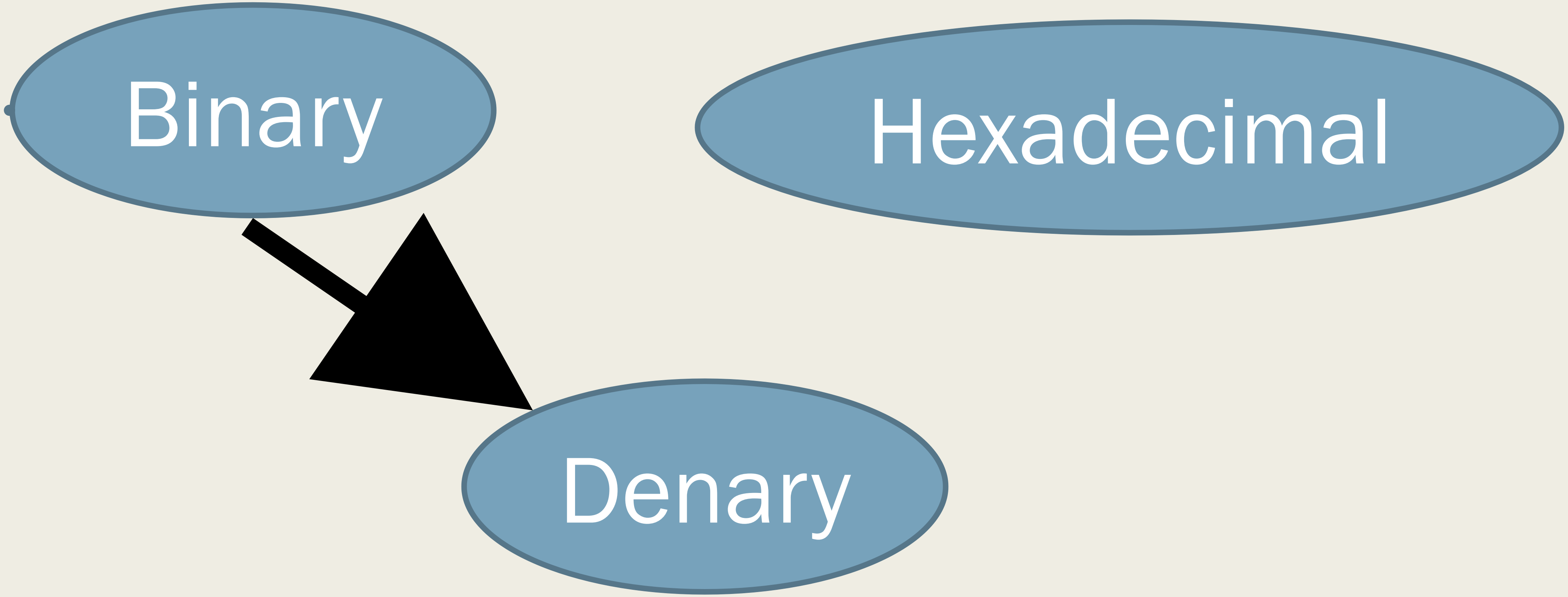
1 2

10^1	10^0
10	1

Tutorial exercise:

- This will provide you with more 4-bit Binary Nibbles to convert to Denary

Number System Triangle



Hexadecimal table:

Denary	Binary				Hexadecimal
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3
4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
9	1	0	0	1	9
10	1	0	1	0	A
11	1	0	1	1	B
12	1	1	0	0	C
13	1	1	0	1	D
14	1	1	1	0	E
15	1	1	1	1	F

Base 10
Denary

17

	10^1	10^0
	10	1

Base 16
Hexadecimal

11

	16^2	16^1	16^0
	256	16	1

Base 10
Denary

1	6	7
10^2	10^1	10^0
100	10	1

Base 16
Hexadecimal

	A	7
16^2	16^1	16^0
256	16	1

In this part we have covered:

- Positional Number Systems
- Positive (unsigned) Integers
  - *Decimal / Denary – Base 10*
  - *Binary - Base 2*
  - *Hexadecimal – Base 16*

In the next part we will cover:

- Converting Denary to Binary

Thank you

These slides have been edited reviewed and amended by Adem Coskun, Izzet Kale and George Charalambous.